

(3 Hours)

Total Marks: 80

Note:

1. **Question No. 1 is compulsory.**
2. Attempt any **THREE** out of the remaining **FIVE** questions.
3. Assume suitable data if necessary.

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|----------|--|-------------|
| 1 | Answer the following (any 4) | (20) |
| | a) Define the terms: Hazard, Vulnerability, Risk | 5 |
| | b) Discuss the Direct and indirect effects of disasters | 5 |
| | c) What is Disaster Scenario of India? | 5 |
| | d) Explain types of Manmade disasters. | 5 |
| | e) What is Climate Change? What are the effects of Global Warming? | 5 |
| 2 | a) What are different types of flood? Enlist structural mitigation measures for flood. | 10 |
| | b) Explain the types of landslide and factors affecting them. Give a case study for the same. | 10 |
| 3 | a) What are different government agencies responsible for various types of disasters? | 10 |
| | b) Explain roles and responsibilities of NDMA in detail. | 10 |
| 4 | a) Discuss the role of GIS and Remote Sensing in disaster management. | 10 |
| | b) Describe the institutional mechanism setup in India. | 10 |
| 5 | a) What is role of NGOs in disaster management? Enlist major NGOs working on disaster management. | 10 |
| | b) Explain Bio shield and Sea wall in detail with schematic diagram. | 10 |
| 6 | a) What is Community Base Disaster Management (CBDM)? Discuss how it is useful in Indian scenario. | 10 |
| | b) What are Do's and don'ts in Earthquake, Tsunami and Cyclone? | 10 |

(3 Hours)

Total Marks: 80

Note:

1. Question No. 1 is compulsory.
2. Attempt any THREE out of the remaining FIVE questions.
3. Assume suitable data if necessary.

Answer the following (any 4) (20)

- a) Define the terms: Mitigation, Triage 5
 - b) What is Climate Change? What are the effects of Global Warming? 5
 - c) What is Disaster Scenario of India? 5
 - d) What are Man-made hazards? Also discuss possibilities of chemical spills. 5
 - e) Explain Risk and Vulnerability. 5
- a) Elaborate the guidelines laid down by NDMA for disaster management in India. 10
 - b) What are different types of flood? Enlist structural mitigation measures for flood. 10
- a) Explain disasters related to industries. Discuss the socioeconomic impact of industrial disasters with a case study. 10
 - b) Enlist and explain various terms and concepts with respect to earthquakes. 10
- a) Explain role of various softwares in disaster management. 10
 - b) What are different mitigation measures for Tsunami and Cyclones? 10
- a) What is the role NGO in disaster management? Enlist major NGOs working on disaster management 10
 - b) Appraise the role of Geo-informatics in disaster management 10
- a) Explain various means of raising finance for mitigating and managing disasters 10
 - b) Discuss the role of GIS and Remote Sensing in disaster management. 10

Hours)

[Total Marks :80]

Notes:

Question No 1 is compulsory.

Answer any 3 from remaining questions.

Illustrate your answers with **neat sketches** wherever necessary.Write proper **Question** and **sub question numbers** as assigned in this question paper.

- 1 Answer any four questions: (20)
 - Justify the significance of studying Disaster Management, highlighting its role in enhancing resilience, minimizing risks, and promoting sustainable development. 5
 - Define and differentiate between Risk and Vulnerability in the context of disaster management. 5
 - Provide an overview of various types of Manmade disasters, elucidating their causes, characteristics, and impacts. 5
 - Analyze the enduring impacts of disasters on affected communities, infrastructure, and socio-economic systems. 5
 - Discuss Climate change, focusing on its underlying causes, ecological repercussions, and implications for disaster risk management. 5
 - Define Bioshield and explore its utility in disaster mitigation, particularly in addressing specific hazards such as floods, landslides, and coastal erosion. 5
- a) Explore the scope and responsibilities of the NIDM in disaster preparedness, training, research, and policy formulation. 10
- b) Describe the Search and Rescue (SAR) procedure in disaster response, supported by a case study illustrating its implementation and effectiveness. 10
- a) Investigate the applications of GIS in disaster management, highlighting its role in spatial analysis, risk assessment, and decision-making. 10
- b) Assess the contributions of various NGOs in Disaster Management, outlining their roles in emergency response, community engagement, and capacity-building initiatives. Additionally, list major NGOs operating globally in this field. 10
- a) Examine the multifaceted components of Disaster Management, including preparedness, response, recovery, and mitigation strategies. 10
- b) Classify different types of droughts and outline structural mitigation measures suitable for drought-prone regions. 10

- Q.5
- a) Evaluate various fundraising mechanisms for disaster management, considering their effectiveness, sustainability, and ethical implications.
 - b) Define Capacity Development and illustrate its application through a case study of Community-Based Disaster Management (CBDM), emphasizing community empowerment, resilience-building, and sustainable development.
- Q.6
- a) Enumerate essential Do's and Don'ts for disaster preparedness and response, providing practical guidelines for individuals and communities to mitigate risks and ensure safety.
 - b) Discuss the roles and functions of various government agencies in Disaster Management in India, emphasizing their coordination and collaboration efforts.
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(3 HOURS)

[Total Marks:80]

B:

Question No:1 is compulsory

Attempt **any three** questions from the remaining **five** questions.

Figures to the right indicates full marks

(20)

Explain PDCA cycle of disaster management.

Define the terms disaster, hazard, vulnerability and risk.

Explain how global warming affects the water cycle and its various components.

Explain disaster management for floods in urban areas.

(20)

What are droughts ? State the mitigation measures taken in drought prone areas.

10

Explain natural and man made disasters. Which disasters can be mitigated and how?

10

(20)

Write a short note on DM Act 2005.

10

Describe in brief about the role of NGO's in disaster management.

05

Explain the concept of Capacity building along with suitable example.

05

(20)

Discuss the applications of remote sensing and GIS in Disaster management process.

10

What are the scope and responsibilities of NIDM and NDMA ?

10

What are the various methods of financial management during a disaster ? Explain the government's fund-raising system for disaster response and recovery.

10

Discuss the role of growing population, industrialization, urbanization, and changing lifestyles in contributing to environmental degradation. What measures can be taken to mitigate their impact?

10

(20)

Write a note on Volcanic eruptions.

08

Explain with examples structural and non-structural mitigation measures

06

What are the dos and don'ts in the event of a Covid like pandemic?

06

Comp / VII / R-19

25/11/24

Time : 3 Hours

Marks : 80

Question no. 1 compulsory.

Answer any three questions out of remaining five

Attempt sub questions in order

Figures to the right indicate full marks.

Explore the following topics (any 4):

[20]

- Evolving approaches to disaster management in India.
- Urban flooding: Implications for major cities.
- Leveraging technology for disaster preparedness.
- Global aid organizations in crisis response.
- Guidelines for earthquake safety.
- Human repercussions of natural calamities.

a) Outline the functions and duties of NIDM

[06]

b) Evaluate the impact of disasters on infrastructure and their hindrance to developmental projects.

[06]

c) Define early warning systems and illustrate their benefits through recent instances of flood, cyclone, and tsunami alerts.

[08]

a) Examine NDMA's directives for disaster preparedness in India.

[06]

b) Examine the collaborative efforts of government bodies and NGOs in disaster management.

[06]

c) Explain Triage. Analyze the importance of initial response in disaster management.

[08]

a) Investigate the contributions of international aid agencies during extreme crises.

[07]

b) Explore the applications of GPS and GIS in drought mitigation strategies.

[07]

c) Describe non-structural measures for pandemic containment.

[06]

a) Assess the involvement of different stakeholders in mass casualty situations.

[06]

b) Delve into the paradox of industrialization: economic progress versus the threat of industrial mishaps.

[08]

c) Highlight the protective role of natural ecosystems against disasters, with examples

[06]

a) Define liquefaction and debate the merits of retrofitting versus relocation in landslide-prone areas.

[06]

b) Define vulnerability and its significance in disaster management.

[04]

c) Enumerate preparedness measures for minimizing chemical disaster losses.

[05]

d) Propose strategies for funding disaster relief efforts and discuss legal considerations

[05]

3 Hours

80 Marks

1. Question no. 1 compulsory.
2. Answer any three questions out of remaining five
3. Attempt sub questions in order
4. Figures to the right indicate full marks.

1. Write short notes on (any 4)

[20]

- a) Capacity building
- b) Functions of NIDM
- c) Sea walls, embankments and bio shields
- d) Triage
- e) Environmental hazard
- f) National Disaster Management Policy
- g) Community based disaster preparedness

2. a) Discuss the framework for disaster management in India.

[8]

2. b) Explain global warming and climate change.

[6]

2. c) Comment on radiation hazards. Also discuss possibilities of chemical spills in Mumbai.

[6]

3. a) Discuss the various types of technological disasters and highlight the specific efforts to mitigate such disasters in India.

[8]

3. b) Explain the role of various international agencies for Disaster Management.

[6]

3. c) Explain various means of raising finance for mitigating and managing disasters

[6]

4. a) Differentiate between structural and non-structural measures of flood mitigation and discuss the importance of forecasting, warning and monitoring system in India

[8]

4. b) Appraise the role of GIS and GPS in disaster management

[6]

4. c) Discuss various types of natural disasters in India and highlight their impacts on life.

[6]

5. a) Explain in detail the design concepts involved in as well as the construction materials used for the safe construction of facilities in case of earthquakes and cyclones. Also discuss the fire resistant facilities that need to be essentially provided in a building/industry.

[8]

5. b) Elaborate the guidelines laid down by NDMA for disaster management in India.

[6]

5. c) Explain in detail, vulnerability, with reference to floods and cyclones. List down the preparatory measures for minimizing vulnerabilities related to Tsunami.

[6]

6. a) Discuss in brief the Disaster Management Act 2005. [6]
6. b) Explain Community based disaster preparedness [5]
6. c) Is rapid depletion of ground water a type of disaster? To which category does this belongs?
What are the reasons for this problem? [5]
6. d) Identify and discuss the various hazards which are associated with volcanic eruptions [4]
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